ML Vision Studio

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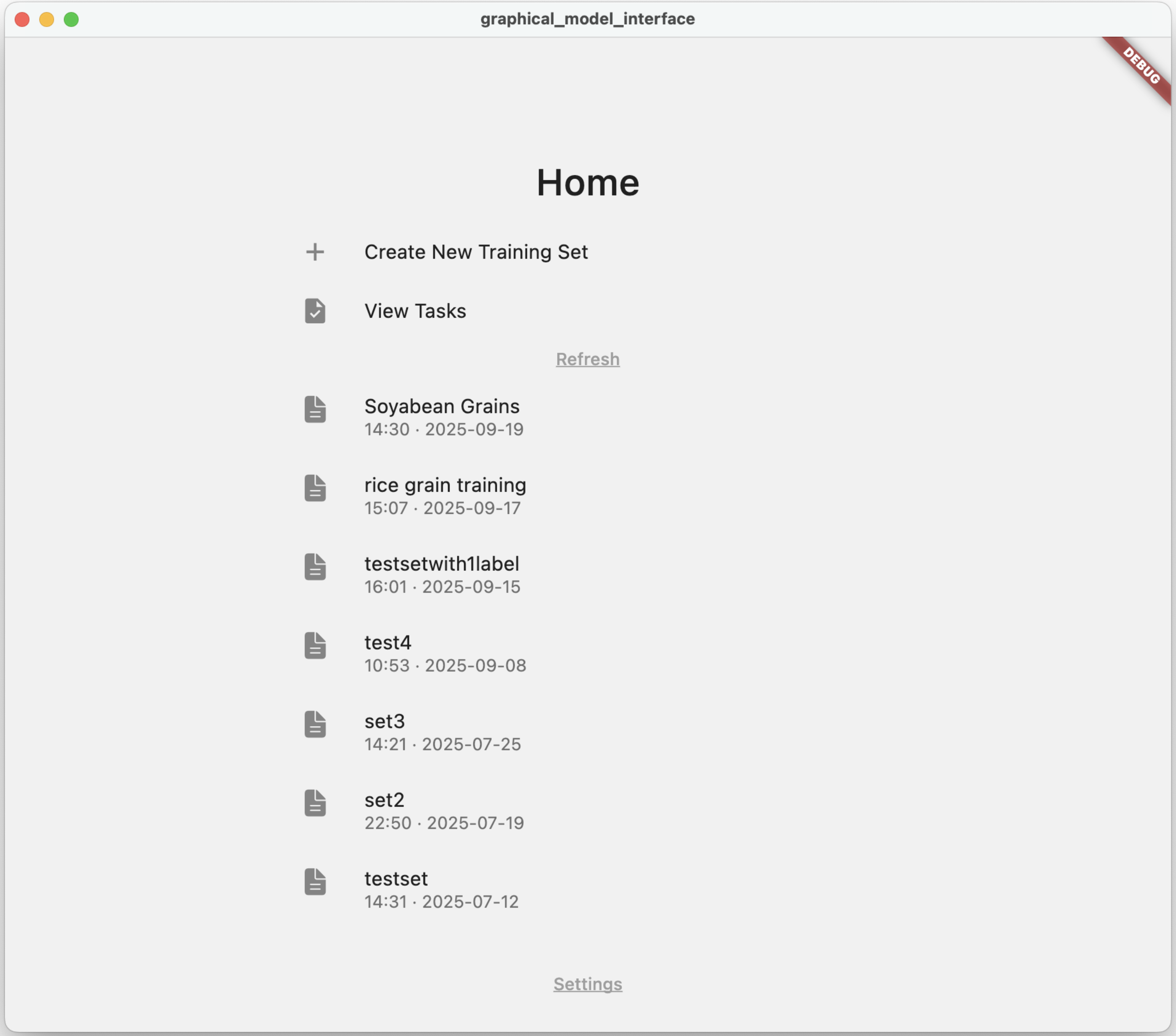
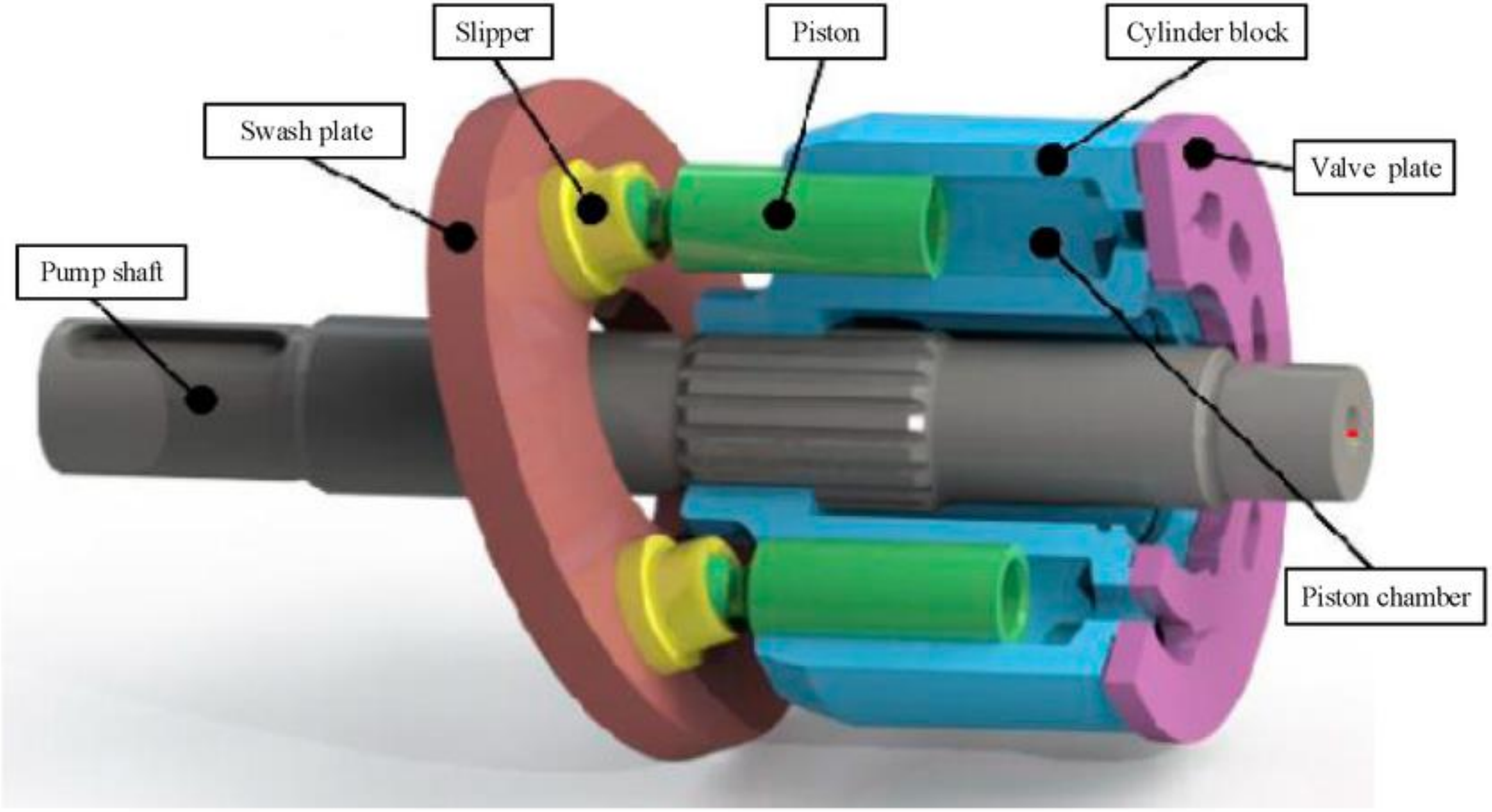
Guided by Prof. Ankur Miglani and Prof. Pavan Kumar Kankar

Last time…


graphical_model_interface
DEBUG
Home
Create New Training Set
View Tasks
Refresh
Soyabean Grains
14:30 · 2025-09-19
rice grain training
15:07 · 2025-09-17
testsetwith1label
16:01 · 2025-09-15
test4
10:53 · 2025-09-08
set3
14:21 · 2025-07-25
set2
22:50 · 2025-07-19
testset
14:31 · 2025-07-12
Settings

Application

Axial Piston Pump – Swash Plate Config

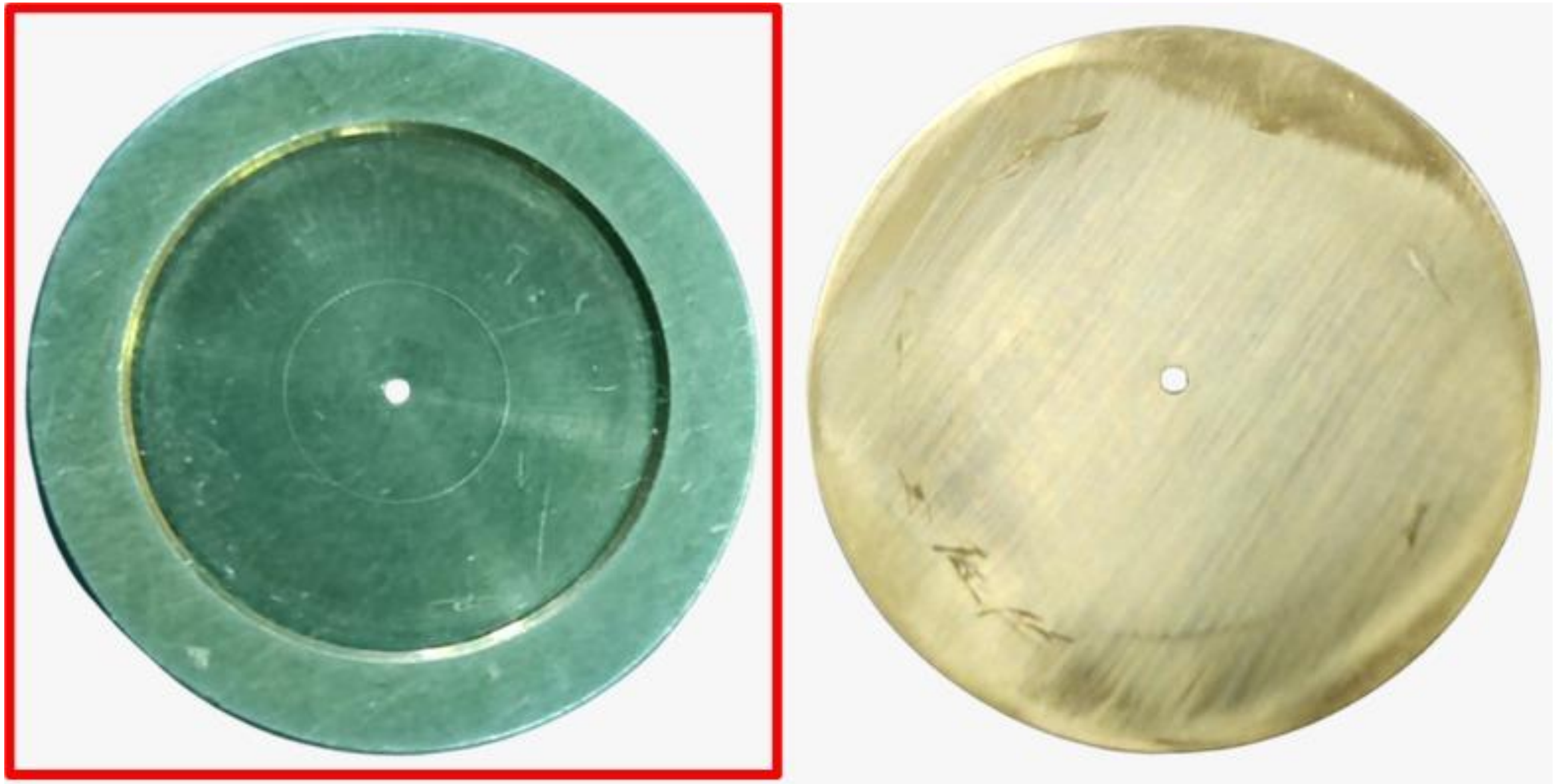


https://www.researchgate.net/figure/Structure-of-Swash-plate-axial-piston-pump_fig1_338483997

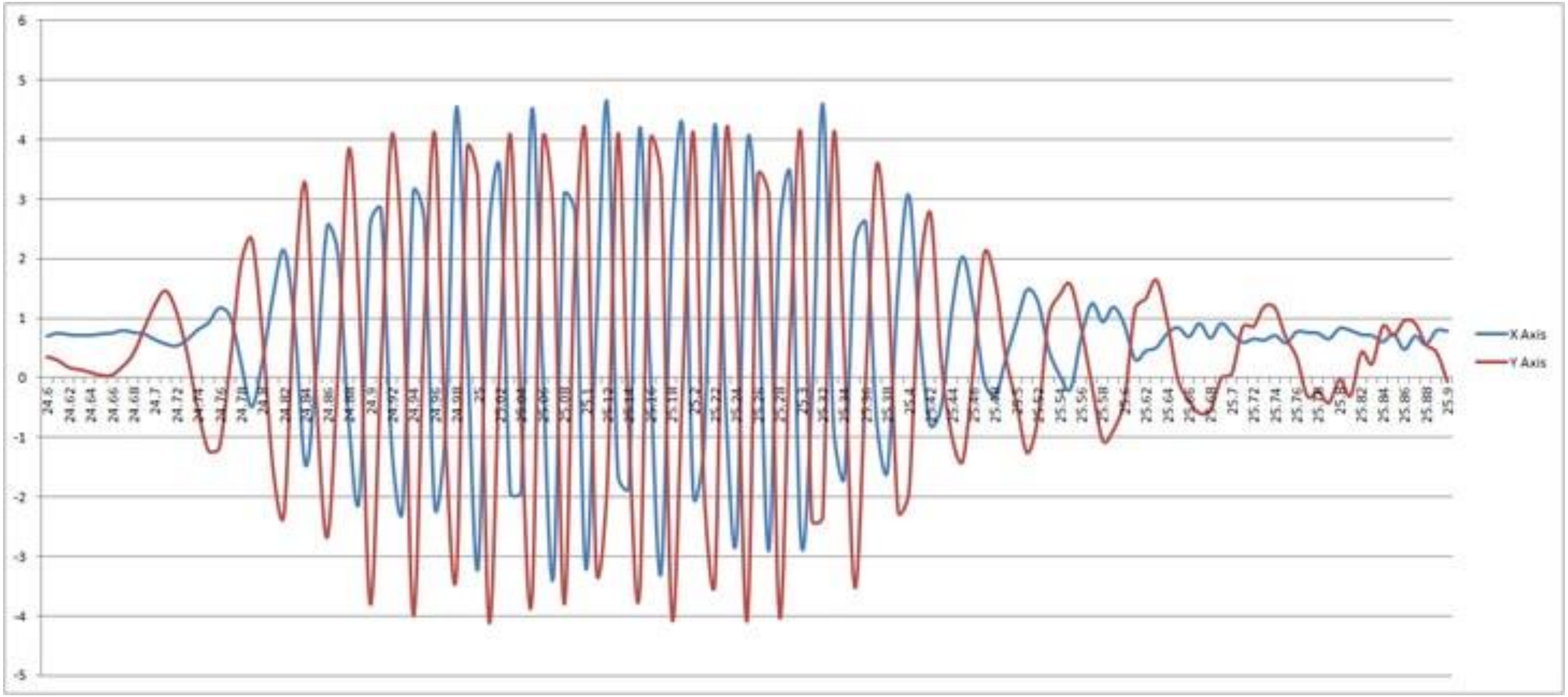
Slippers

https://www.powermotiontech.com/hydraulics/hydraulic-pumps-motors/article/21136312/common-modes-of-failure-for-hydraulic-piston-pumps

Slipper wear



(c) (d)


X Axis
Y Axis


X
Y

Levels of wear

1. Healthy
2. 10 μm
3. 45 μm
4. 75 μm
5. 100 μm

5 classes for prediction

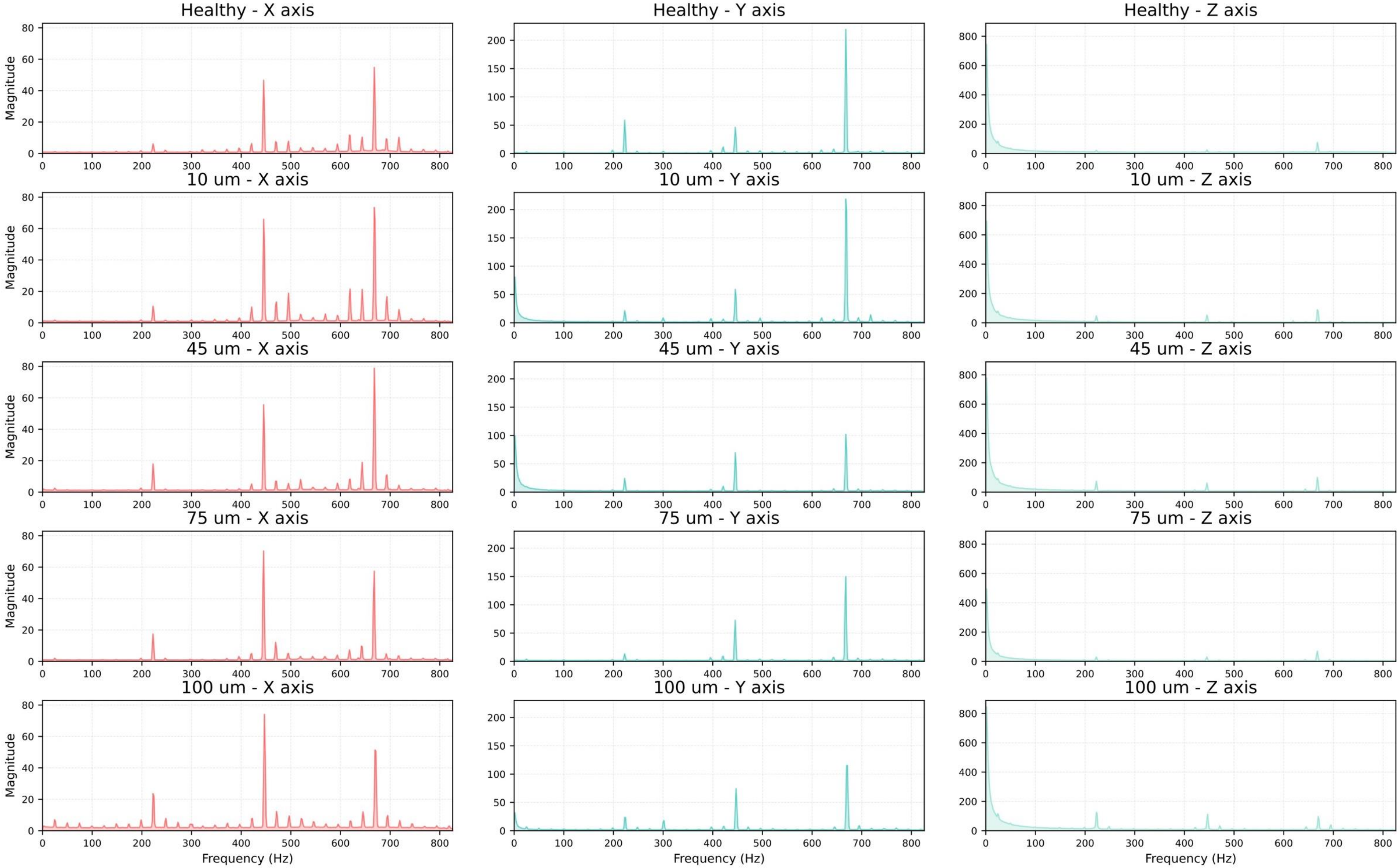
Vibration records

Channel name	X	Y	Z
0.0	-0.545170038	9.759454571	5.147208811
0.0006054999999776	3.726288255	-13.04648045	-5.812913371
0.0012109999999836	-4.380038743	1.257279966	1.87762764
0.0018163999999956	-4.793954241	2.785649131	1.184890427
0.0024219000000016	10.90139041	2.527377161	-3.731239836
0.0030273999999792	-4.107949672	-0.381909252	2.825112607
0.0036327999999912	-1.721388414	-6.004495831	-3.180082208
0.0042382999999972	3.441161583	14.67644444	4.987961621
0.0048438000000032	1.513523891	-5.85785282	-0.672356783
0.0054491999999868	1.26172813	-7.591683982	-2.119667825
0.0060546999999928	-9.64428203	5.427996729	4.028061581
0.0066601999999988	9.450815553	0.545674241	-4.613426292

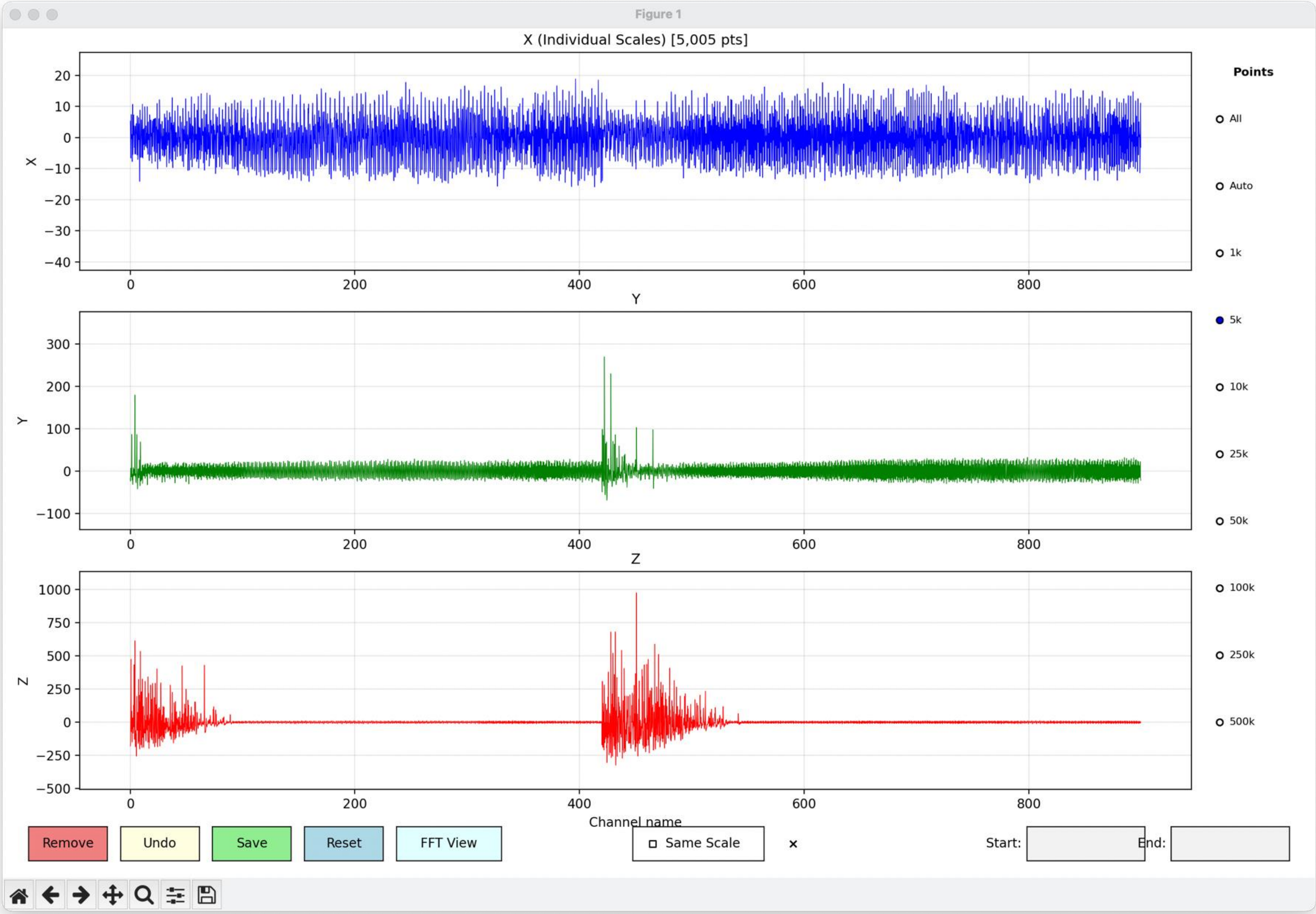
- Channel name – time
- X – acceleration in x direction
- Y – acceleration in y direction
- Z – acceleration in z direction


Vibration Data Analysis - Time Domain
Healthy - X axis
Healthy - Y axis
Healthy - Z axis
10 um - X axis
10 um - Y axis
10 um - Z axis
45 um - X axis
45 um - Y axis
45 um - Z axis
75 um - X axis
75 um - Y axis
75 um - Z axis
100 um - X axis
100 um - Y axis
100 um - Z axis
Acceleration
Time (s)

Vibration Data Analysis - Frequency Domain (FFT)



Removing transients



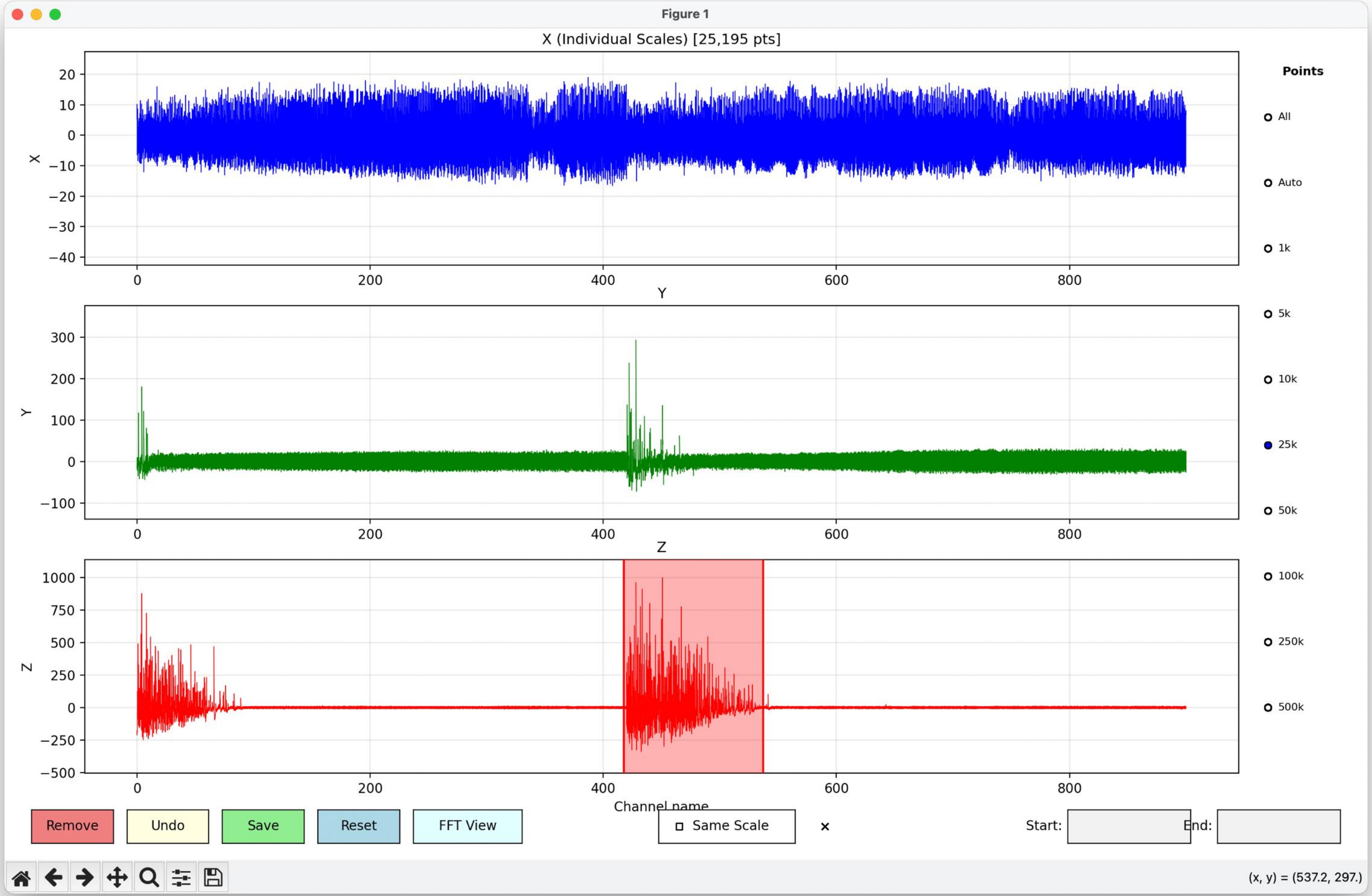


Figure 1
X (Individual Scales) [25,195 pts]
X
Y
Z
Channel name
Points
All
Auto
1k
5k
10k
25k
50k
100k
250k
500k
Remove
Undo
Save
Reset
FFT View
Same Scale
Start:
End:
(x, y) = (537.2, 297.)

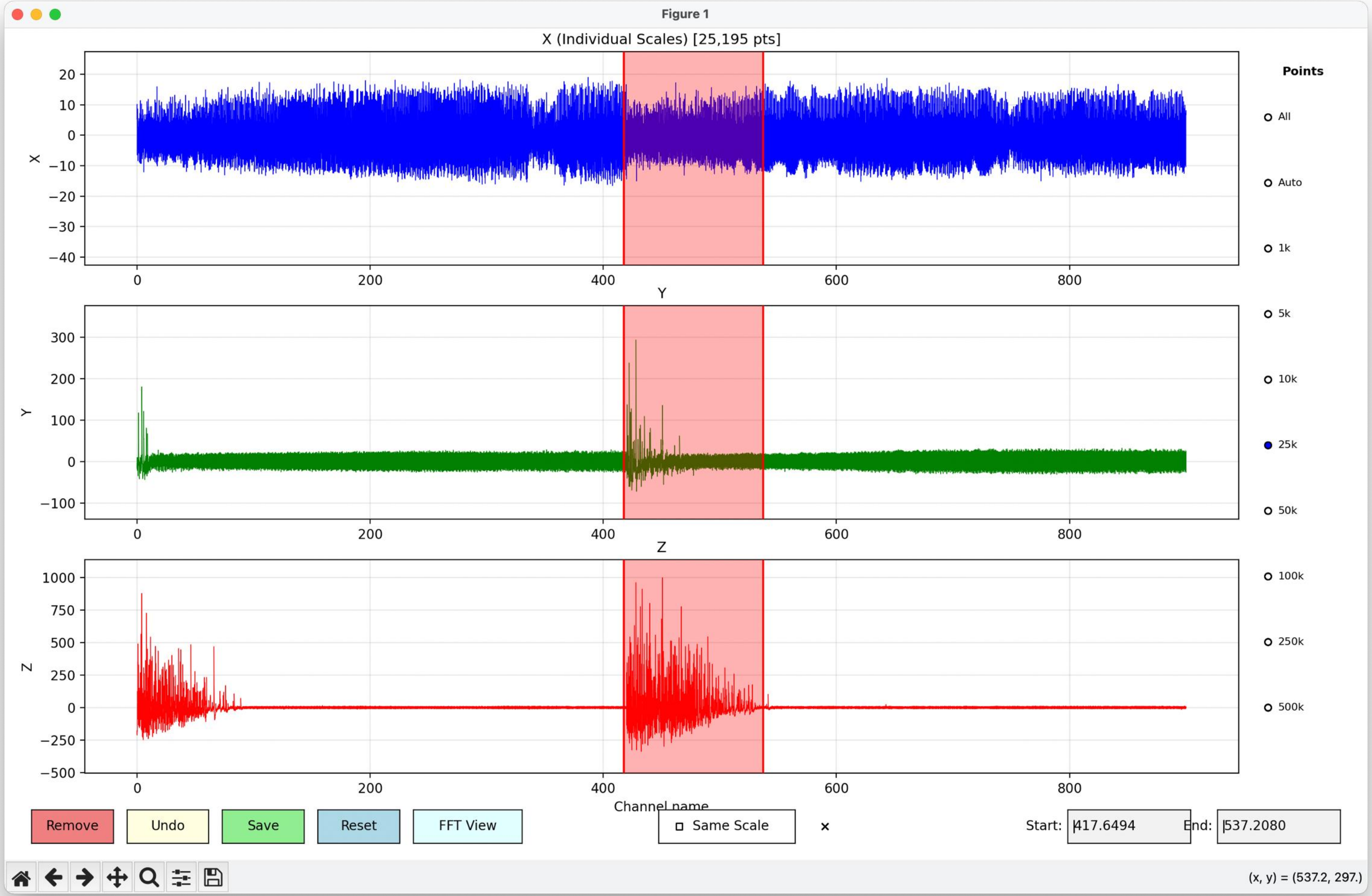


Figure 1
X (Individual Scales) [25,195 pts]
Points
All
Auto
1k
5k
10k
25k
50k
100k
250k
500k
X
Y
Z
Channel name
Remove
Undo
Save
Reset
FFT View
Same Scale
Start: 417.6494
End: 537.2080
(x, y) = (537.2, 297.)

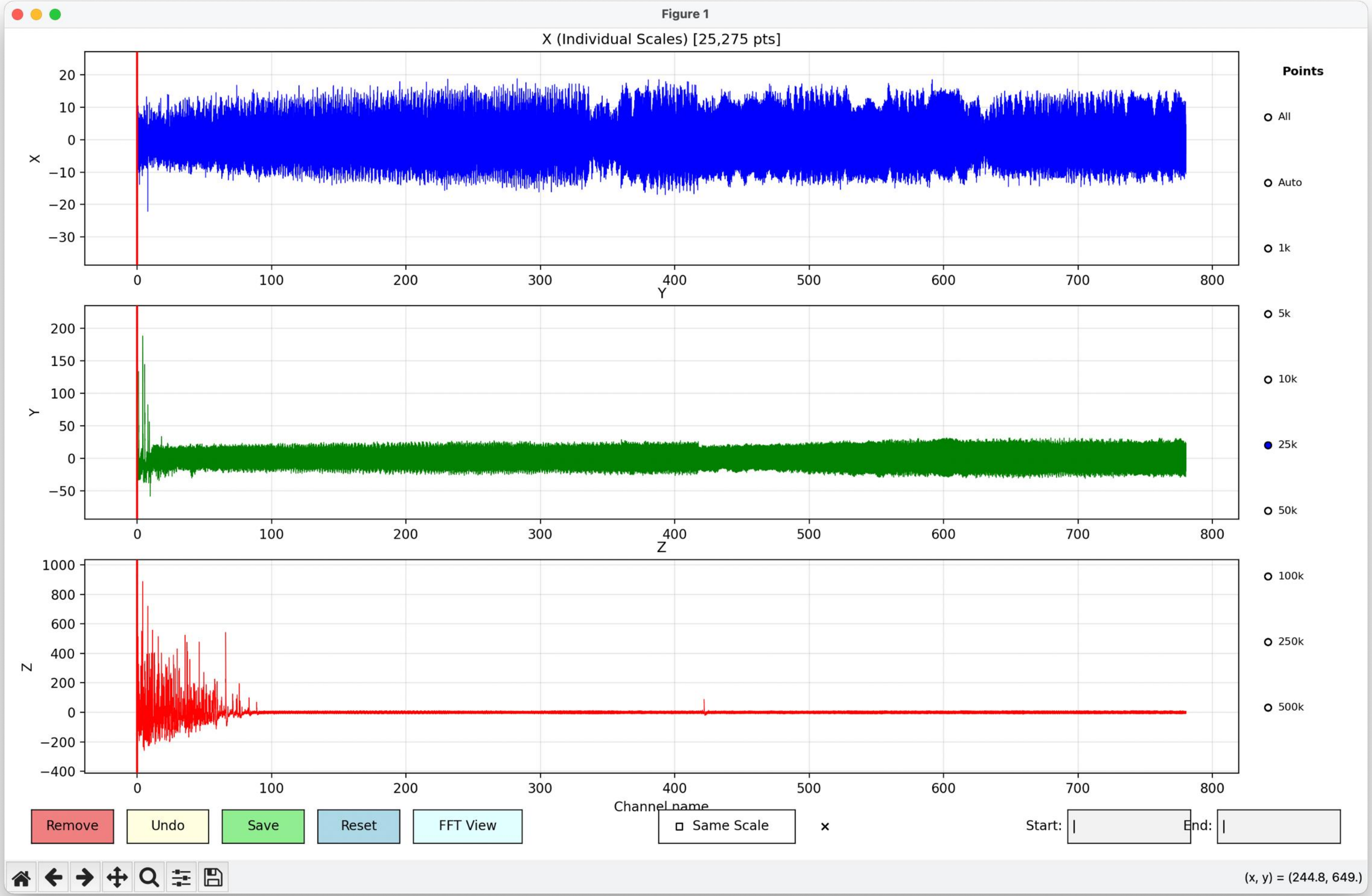
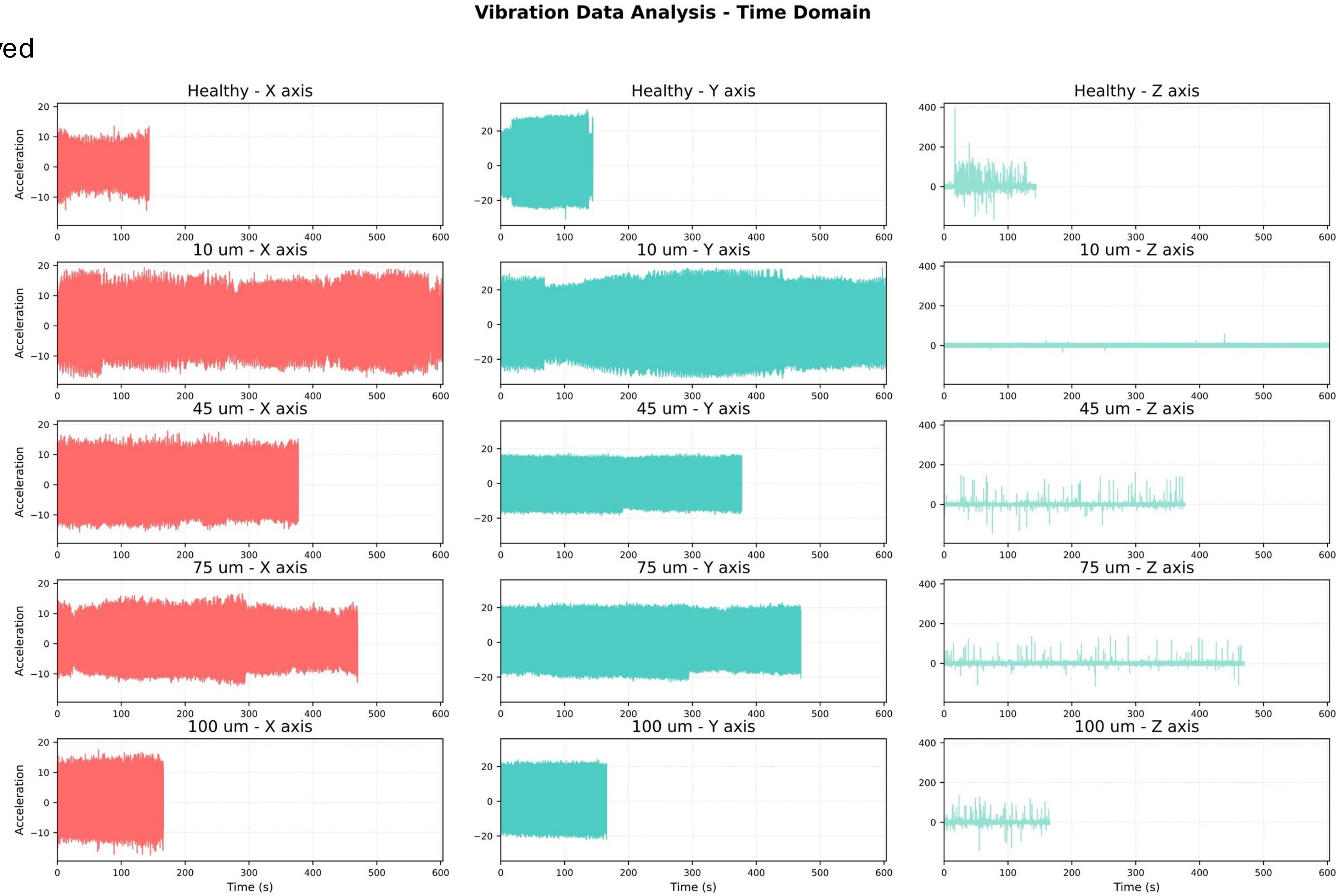


Figure 1
X (Individual Scales) [25,275 pts]
X
Y
Z
Channel name
Points
All
Auto
1k
5k
10k
25k
50k
100k
250k
500k
Remove
Undo
Save
Reset
FFT View
Same Scale
Start:
End:
(x, y) = (244.8, 649.)

Figure 1
X FFT (Welch) (Original Data) (Individual Scales, Log)
PSD [V²/Hz]
Y FFT (Welch) (Original Data)
PSD [V²/Hz]
Z FFT (Welch) (Original Data)
PSD [V²/Hz]
Frequency [Hz]
Undo
Save
Reset
Time View
Current
Same Scale
Log Scale
(x, y) = (674.2, 0.00338411)

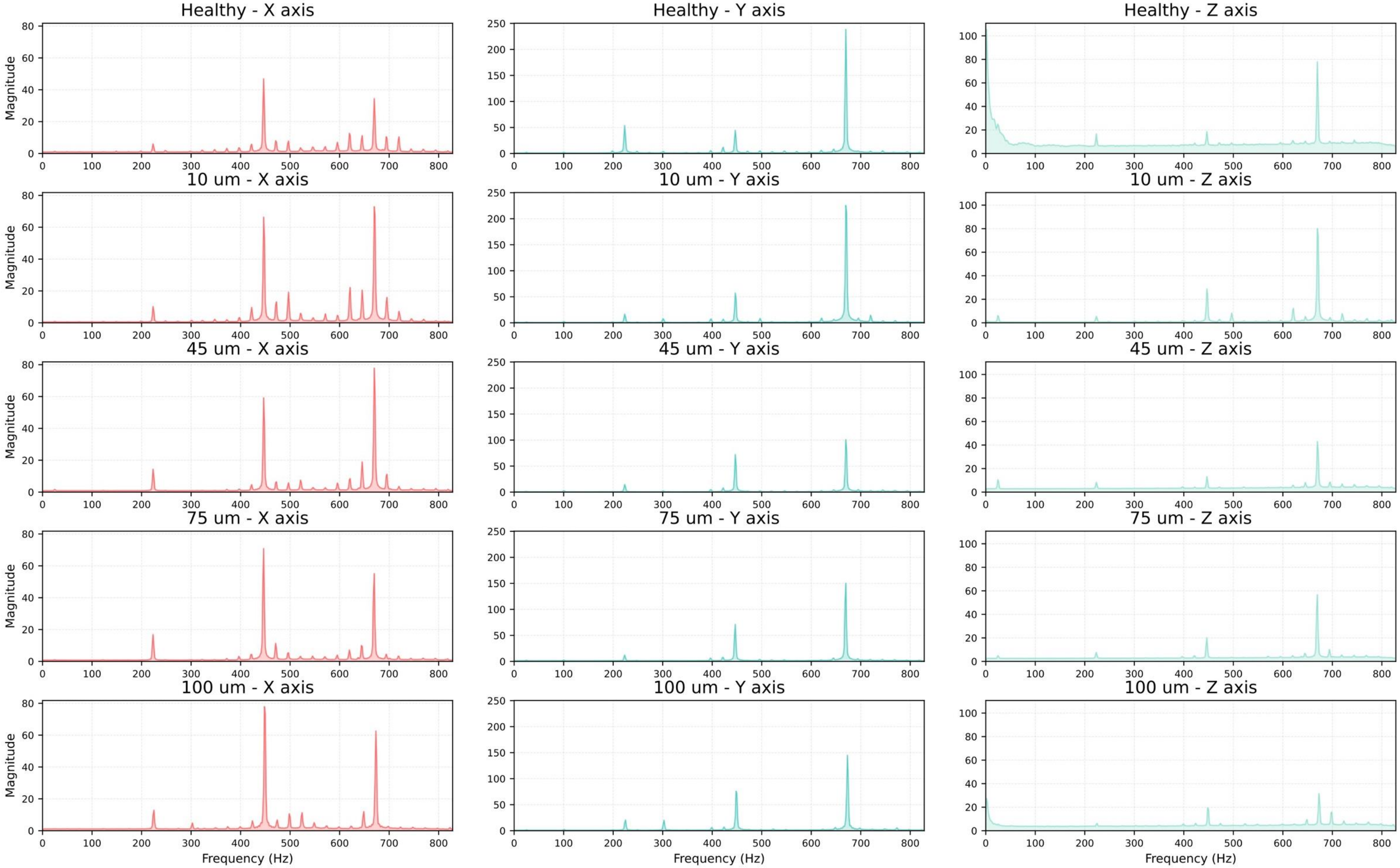

Figure 1
X FFT (Welch) (Current Data) (Individual Scales, Log)
Y FFT (Welch) (Current Data)
Z FFT (Welch) (Current Data)
PSD [V²/Hz]
Frequency [Hz]
Undo
Save
Reset
Time View
Original
Same Scale
Log Scale

Removed


Vibration Data Analysis - Time Domain
Healthy - X axis
Healthy - Y axis
Healthy - Z axis
10 um - X axis
10 um - Y axis
10 um - Z axis
45 um - X axis
45 um - Y axis
45 um - Z axis
75 um - X axis
75 um - Y axis
75 um - Z axis
100 um - X axis
100 um - Y axis
100 um - Z axis
Acceleration
Time (s)

Vibration Data Analysis - Frequency Domain (FFT)

Removed

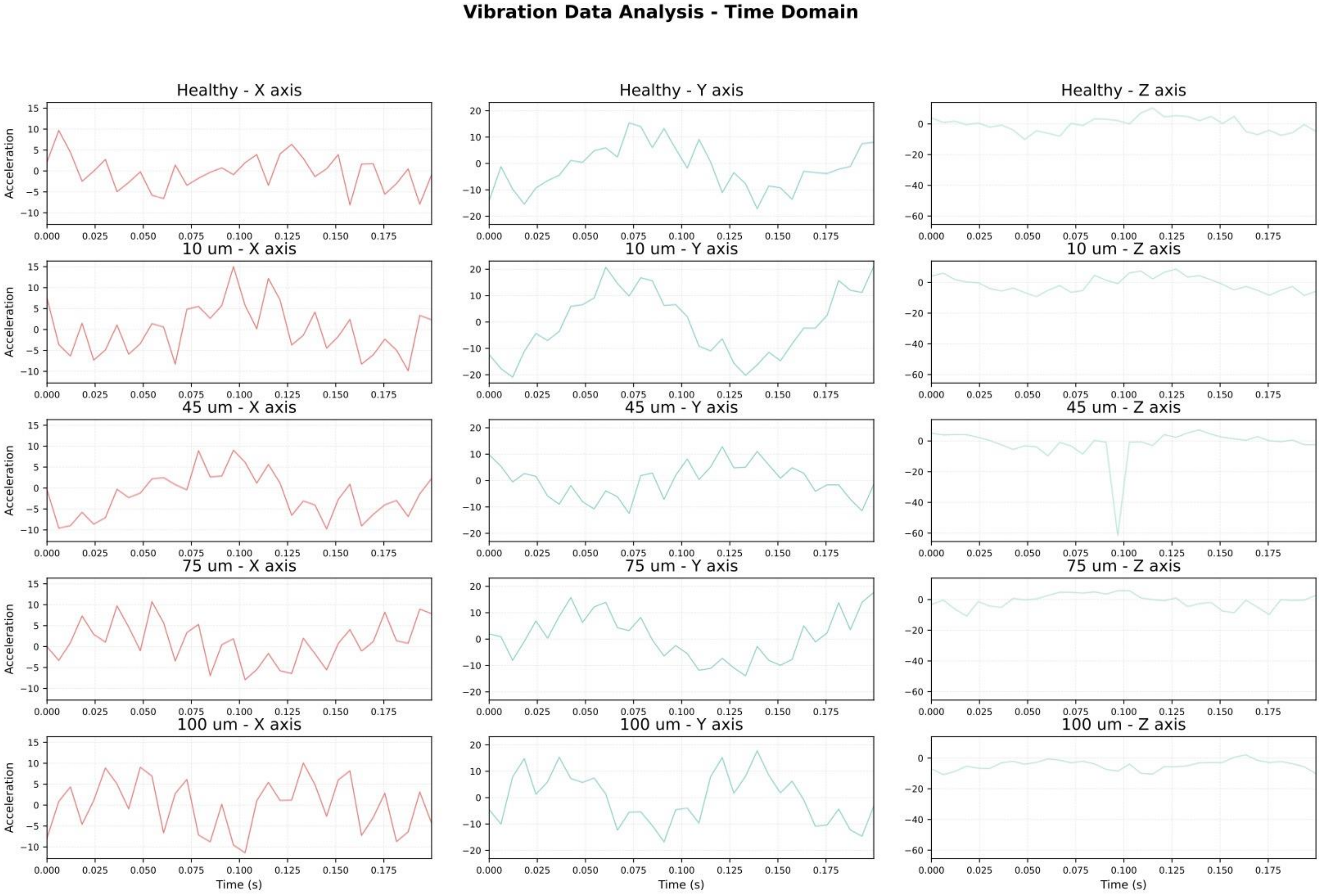


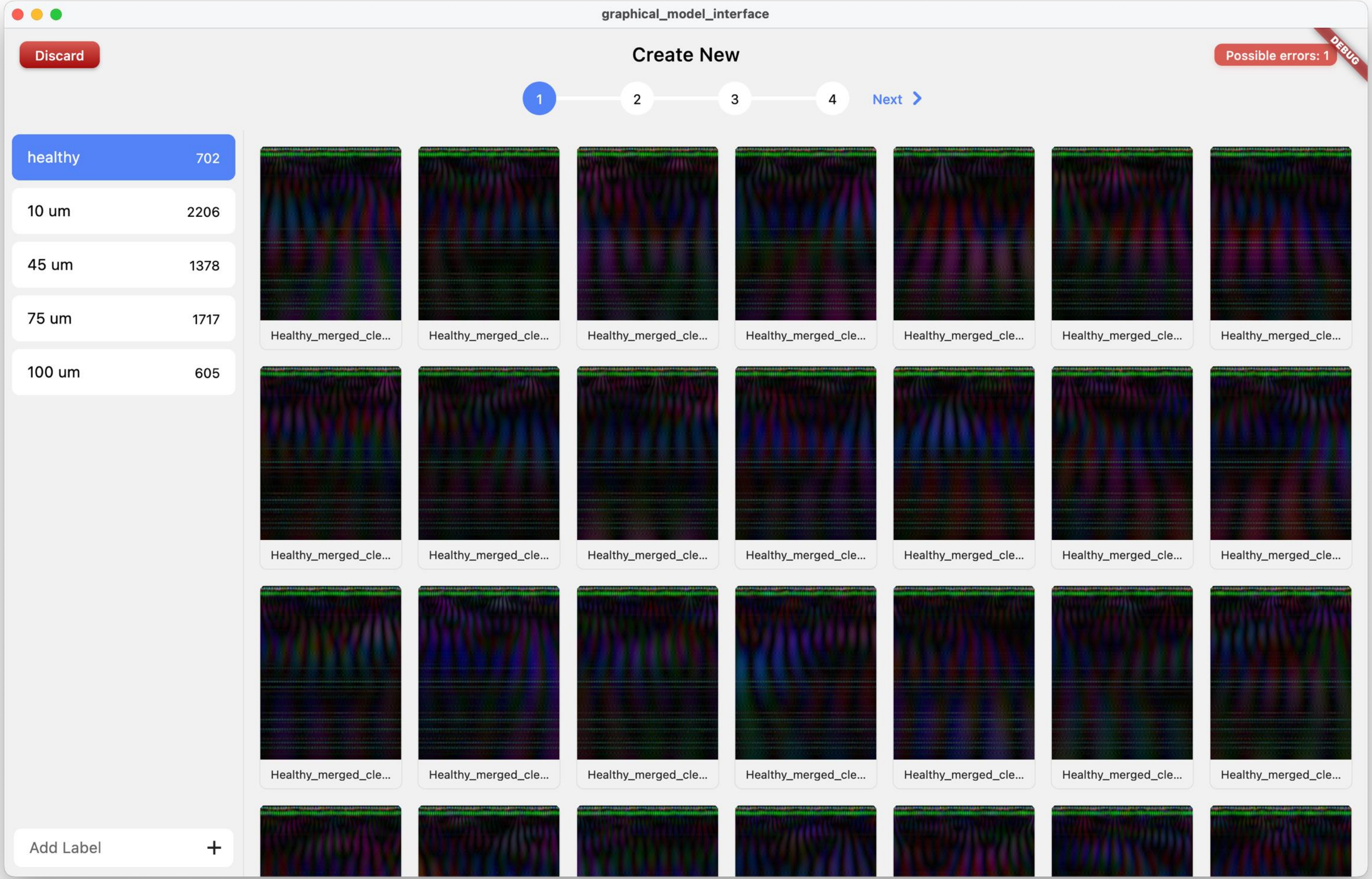
Original

Vibration Data Analysis - Frequency Domain (FFT)
Healthy - X axis
Healthy - Y axis
Healthy - Z axis
10 um - X axis
10 um - Y axis
10 um - Z axis
45 um - X axis
45 um - Y axis
45 um - Z axis
75 um - X axis
75 um - Y axis
75 um - Z axis
100 um - X axis
100 um - Y axis
100 um - Z axis
Magnitude
Frequency (Hz)

Breaking records into small pieces

- Broke each recording of each class into chunks of equal size
- Each chunk lasts for ~ 0.2 seconds
- Each chunk has 340 points




graphical_model_interface
Discard
Create New
Possible errors: 1
DEBUG
1
2
3
4
Next
healthy 702
10 um 2206
45 um 1378
75 um 1717
100 um 605
Add Label
Healthy_merged_cle...

CNN models?

ResNet

MobileNetV2

MobileNetV3

InceptionV3

EfficientNet

Custom

CNN models?

ResNet

MobileNetV2

MobileNetV3

InceptionV3

EfficientNet

Custom

Layer (type)	Output Shape	Param # [Total 102,725 (401.27 KB)]
input_layer (InputLayer)	(None, 256, 256, 3)	0
sequential (Sequential)	(None, 256, 256, 3)	0
rescaling (Rescaling)	(None, 256, 256, 3)	0
conv2d (Conv2D)	(None, 256, 256, 32)	896
batch_normalization (BatchNormalization)	(None, 256, 256, 32)	128
re_lu (ReLU)	(None, 256, 256, 32)	0
max_pooling2d (MaxPooling2D)	(None, 128, 128, 32)	0
conv2d_1 (Conv2D)	(None, 128, 128, 64)	18496
batch_normalization_1 (BatchNormalization)	(None, 128, 128, 64)	256
re_lu_1 (ReLU)	(None, 128, 128, 64)	0
max_pooling2d_1 (MaxPooling2D)	(None, 64, 64, 64)	0
conv2d_2 (Conv2D)	(None, 64, 64, 128)	73856
batch_normalization_2 (BatchNormalization)	(None, 64, 64, 128)	512
re_lu_2 (ReLU)	(None, 64, 64, 128)	0
max_pooling2d_2 (MaxPooling2D)	(None, 32, 32, 128)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 128)	0
dense (Dense)	(None, 64)	8256
dropout (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 5)	325

Performance

$$\begin{bmatrix} 441 & 0 & 0 & 0 & 0 \\ 0 & 121 & 0 & 0 & 0 \\ 0 & 0 & 275 & 0 & 0 \\ 0 & 0 & 0 & 341 & 3 \\ 0 & 0 & 0 & 0 & 106 \end{bmatrix}$$

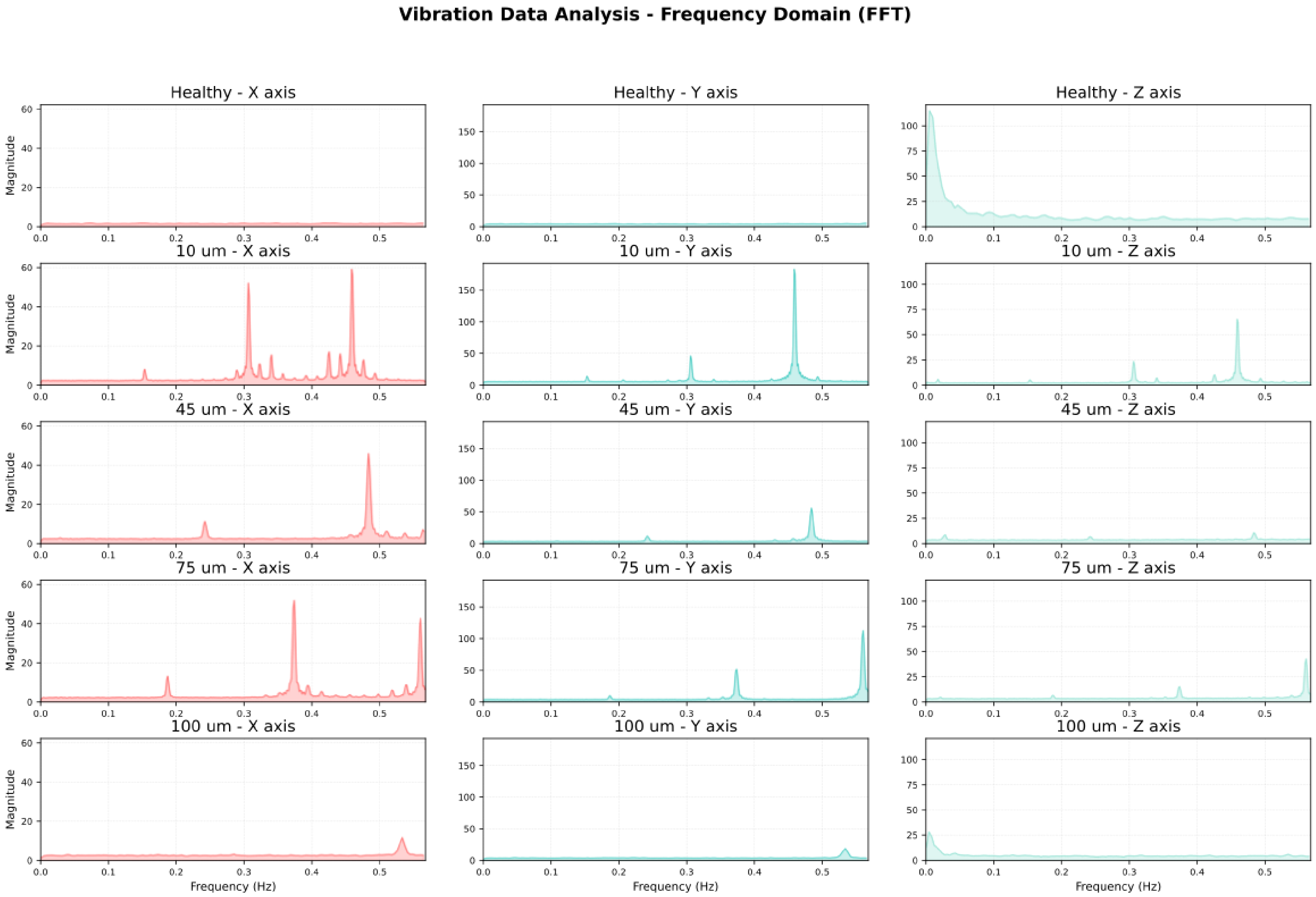
	precision	recall	f1-score	support
10 um	1.00	1.00	1.00	441
100 um	1.00	1.00	1.00	121
45 um	1.00	1.00	1.00	275
75 um	1.00	0.99	1.00	344
Healthy	0.97	1.00	0.99	106
accuracy			1.00	1287
macro avg	0.99	1.00	1.00	1287
weighted avg	1.00	1.00	1.00	1287

Test Loss: 0.012958236038684845
Test Accuracy: 0.997668981552124

$$Precision = \frac{TP}{TP + FP} \qquad Recall = \frac{TP}{TP + FN} \qquad F1 = \frac{2 * P * R}{P + R}$$

99.7% accuracy? Test further...

- Added Gaussian Noise to Test data

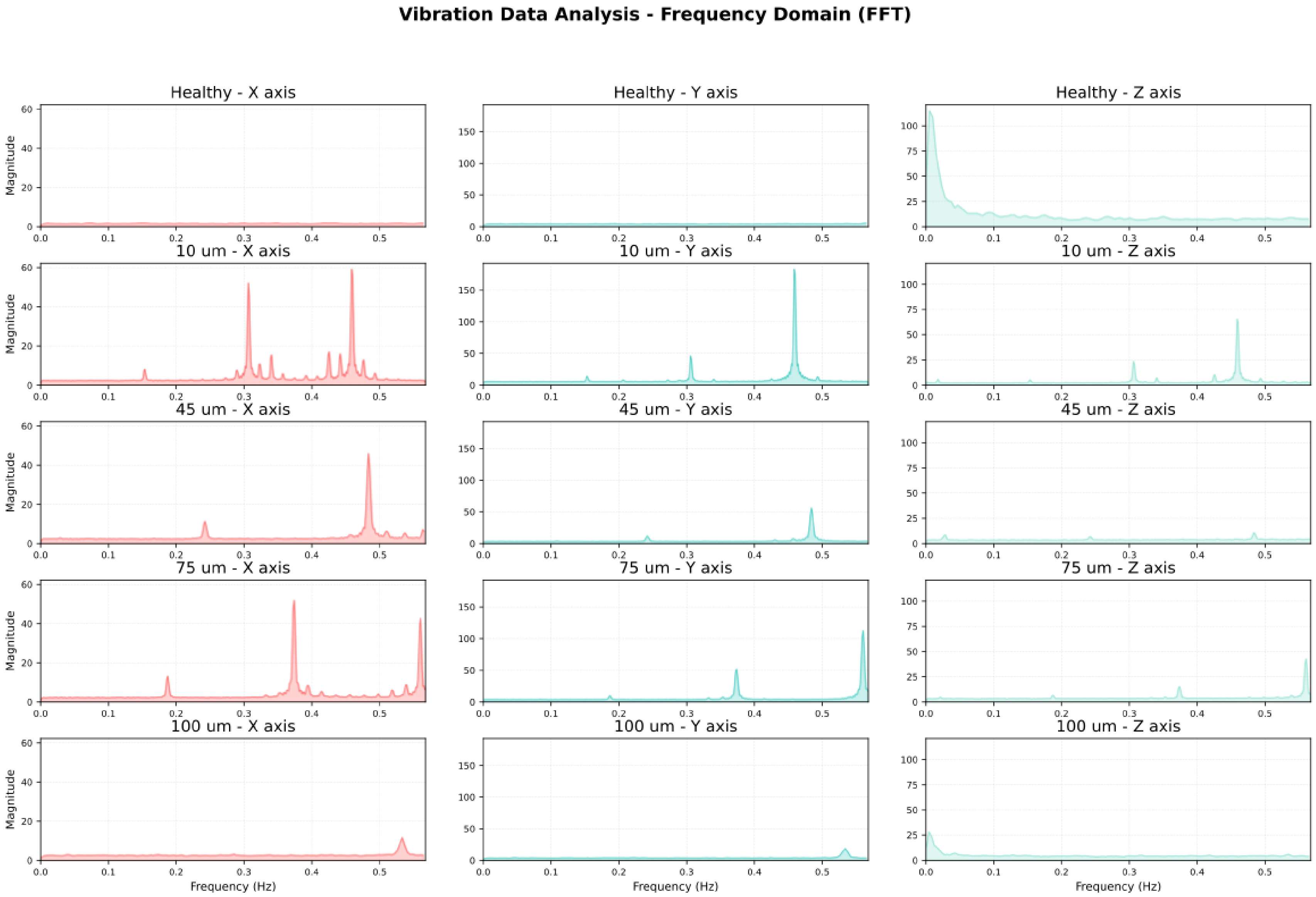


99.7% accuracy? Test further...

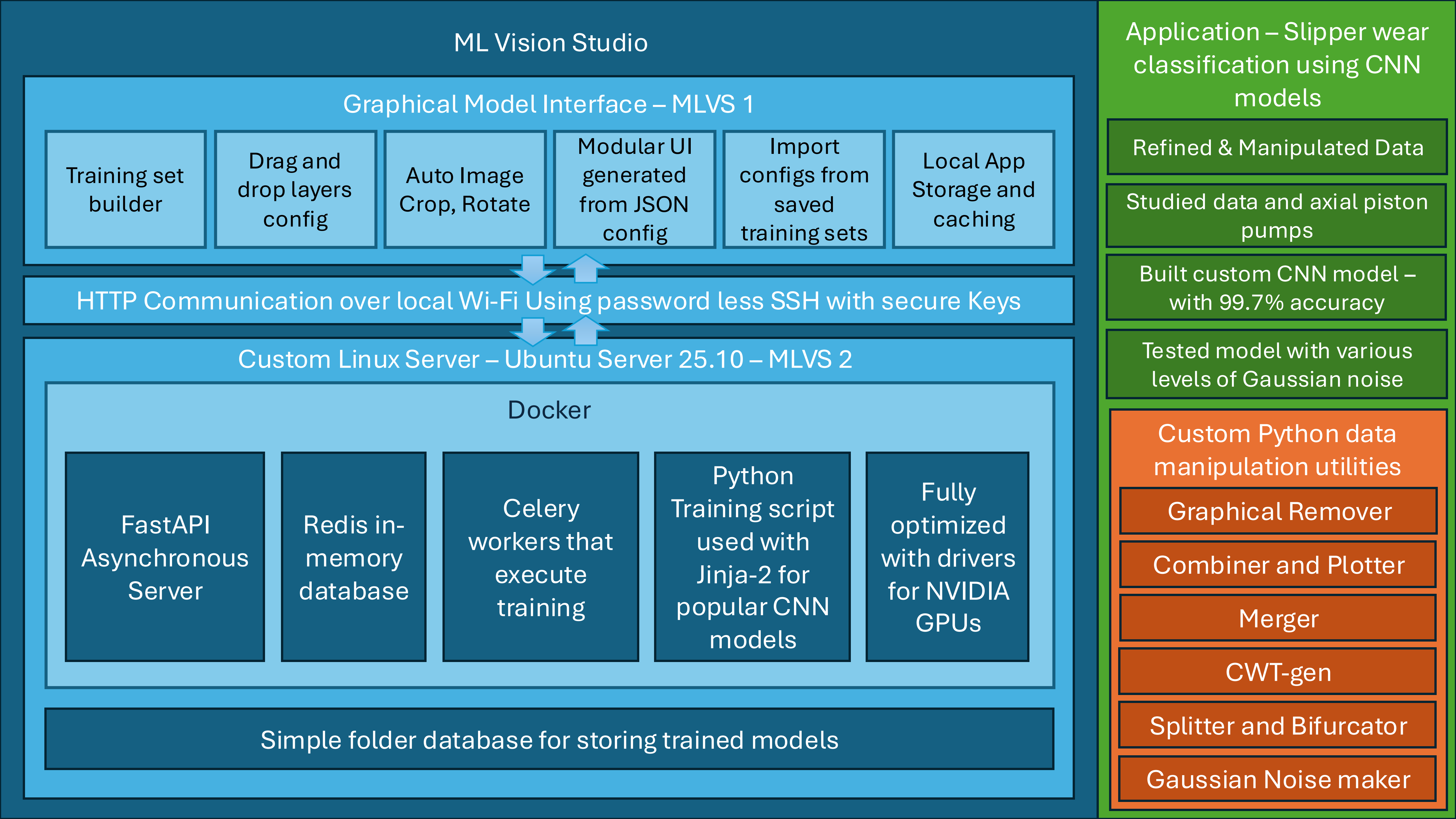
- Added Gaussian Noise to Test data

Accuracy:

- 30 db– 98.7%
- 15 db – 69.4%
- 10 db – 7%



Summary


ML Vision Studio
Graphical Model Interface – MLVS 1
Training set builder
Drag and drop layers config
Auto Image Crop, Rotate
Modular UI generated from JSON config
Import configs from saved training sets
Local App Storage and caching
HTTP Communication over local Wi-Fi Using password less SSH with secure Keys
Custom Linux Server – Ubuntu Server 25.10 – MLVS 2
Docker
FastAPI Asynchronous Server
Redis in-memory database
Celery workers that execute training
Python Training script used with Jinja-2 for popular CNN models
Fully optimized with drivers for NVIDIA GPUs
Simple folder database for storing trained models
Application – Slipper wear classification using CNN models
Refined & Manipulated Data
Studied data and axial piston pumps
Built custom CNN model – with 99.7% accuracy
Tested model with various levels of Gaussian noise
Custom Python data manipulation utilities
Graphical Remover
Combiner and Plotter
Merger
CWT-gen
Splitter and Bifurcator
Gaussian Noise maker

Domains

- Linux Server Systems
- Remote management SSH
- Software engineering
- Machine Learning
- Data Science
- Computer Networking

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